

Optimal Real-Data Allocation for Synthetic-Data-Augmented Inference

Setup, Model, and Theory

1 Setup and Model

1.1 Probability space, estimand, and observables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a measurable space $(\mathcal{X}, \mathcal{B})$. Let P^* denote the data-generating distribution on $\mathcal{X}$ and let $\theta : \mathcal{P} \rightarrow \mathbb{R}$ be a real-valued functional defined on a class $\mathcal{P}$ of distributions on $\mathcal{X}$ that contains P^* and the synthetic distributions introduced below. The target estimand is $\theta^* := \theta(P^*)$. We treat θ^* as scalar throughout this section; the vector extension is recorded in Remark 1.3.

The researcher observes a real sample

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P^*. \quad (1)$$

She also has access to a calibratable synthetic-data generator, formalized below. The synthetic generator's randomness is driven by an auxiliary i.i.d. stream $\{U_j\}_{j \geq 1}$ in some space $(\mathcal{U}, \mathcal{B}_{\mathcal{U}})$ with distribution Q , independent of $\{X_i\}$. Sample-size sequences $m_n, x, n_v \in \mathbb{N}$ index the synthetic, calibration, and validation budgets respectively.

1.2 Sample-splitting partition

Let $\Pi_n = (C_x, E_x, V_x)$ denote a measurable partition of $[n] := \{1, \dots, n\}$ into a *calibration* subset C_x of size x , an *estimation* subset E_x , and a (possibly empty) *validation* subset V_x . Two configurations are used:

- **Linear configuration.** $V_x = \emptyset$ and $E_x = [n] \setminus C_x$, so $|E_x| = n - x$.
- **PPI configuration.** $E_x = \emptyset$ and $V_x = [n] \setminus C_x$, so $|V_x| = n - x$.

Sample splitting is enforced throughout: the estimators we construct depend on the partition only through the indicated subsets, and the partition is independent of $\{X_i\}$ (e.g. a fixed deterministic partition or one drawn from an independent randomization device). Cross-fitting averages over an ensemble of partitions and is treated in the algorithmic section. Let $\mathcal{F}_C := \sigma(\{X_i\}_{i \in C_x})$ denote the calibration σ -field.

1.3 Real-data estimator

Assumption 1.1 (Asymptotic linearity of the real estimator). There exists a measurable function $\psi : \mathcal{X} \rightarrow \mathbb{R}$ with $\mathbb{E}_{P^*}[\psi(X_1)] = 0$ and $\text{Var}_{P^*}(\psi(X_1)) = a \in (0, \infty)$ such that, for every deterministic sequence of estimation subsets $E \subset [n]$ with $|E| \rightarrow \infty$,

$$\hat{\theta}_R(E) - \theta^* = \frac{1}{|E|} \sum_{i \in E} \psi(X_i) + r_R(E), \quad (2)$$

where the remainder satisfies $\mathbb{E}[r_R(E)] = 0$ and $\mathbb{E}[r_R(E)^2] = o(|E|^{-1})$.

The function ψ is the influence function of $\hat{\theta}_R$. Under Assumption 1.1,

$$V_R(x) := \text{Var}(\hat{\theta}_R(E_x)) = \frac{a}{n-x} + o((n-x)^{-1}), \quad |E_x| = n-x. \quad (3)$$

We write $V_R(x) = a/(n-x)$ for the leading-order variance, absorbing the $o(\cdot)$ remainder into the $1 + o(1)$ factors that appear throughout.

1.4 Calibratable synthetic generator

A *calibration channel* is a measurable family of stochastic kernels

$$\mathcal{G} : \mathcal{X}^x \times \mathcal{U} \rightarrow \mathcal{X}, \quad \mathcal{G}(\mathbf{X}_C; U) \sim P_{\mathbf{X}_C} \quad \text{for } \mathbf{X}_C \in \mathcal{X}^x, U \sim Q, \quad (4)$$

inducing a calibrated synthetic distribution $P_x := P_{\mathbf{X}_C}$ that depends on the calibration sample $\mathbf{X}_C = \{X_i\}_{i \in C_x}$. The pseudo-true parameter is $\theta_x := \theta(P_x)$, which is $\mathcal{F}_C$ -measurable and in general nondegenerate. The *calibration error* is

$$\Delta(x) := \theta_x - \theta^*, \quad \mathbb{E}[\Delta(x) \mid \mathcal{F}_C] = \Delta(x). \quad (5)$$

Define the *expected squared calibration error* as

$$b^2(x) := \mathbb{E}[\Delta(x)^2] = (\mathbb{E}[\Delta(x)])^2 + \text{Var}(\Delta(x)). \quad (6)$$

This is the unconditional second moment of the bias and decomposes into systematic and stochastic components; both are absorbed into the working model below. (The previous draft conflated the random $\Delta(x)$ with its L^2 -norm; the working assumption is on the deterministic quantity $b^2(x)$.)

1.5 Synthetic estimator

Conditionally on $\mathcal{F}_C$, draw synthetic observations

$$Z_1, \dots, Z_{m_n} \mid \mathcal{F}_C \stackrel{\text{i.i.d.}}{\sim} P_x, \quad (7)$$

generated using fresh randomness from the auxiliary stream and independent of $\hat{\theta}_R(E_x)$. The synthetic estimator $\hat{\theta}_S(x)$ is computed from $\{Z_j\}_{j=1}^{m_n}$.

Assumption 1.2 (Asymptotic linearity of the synthetic estimator). There exists a measurable family $(\phi_x)_{x \geq 1}$ such that conditionally on $\mathcal{F}_C$,

$$\hat{\theta}_S(x) - \theta_x = \frac{1}{m_n} \sum_{j=1}^{m_n} \phi_x(Z_j) + r_S(x), \quad (8)$$

with $\mathbb{E}[\phi_x(Z_1) \mid \mathcal{F}_C] = 0$, $\text{Var}(\phi_x(Z_1) \mid \mathcal{F}_C) = \sigma_S^2(x)$, $\mathbb{E}[r_S(x) \mid \mathcal{F}_C] = 0$, and $\mathbb{E}[r_S(x)^2] = o(m_n^{-1})$. Furthermore there exists a constant $\bar{\sigma}_S^2 < \infty$ such that $\mathbb{E}[\sigma_S^2(x)] \rightarrow \bar{\sigma}_S^2$ uniformly across the regimes considered.

Define the synthetic sampling variance

$$v_n := \bar{\sigma}_S^2/m_n, \quad \mathbb{E}[\text{Var}(\hat{\theta}_S \mid \mathcal{F}_C)] = v_n\{1 + o(1)\}. \quad (9)$$

Lemma 1.1 (Synthetic-error decomposition). *Under Assumption 1.2,*

$$\mathcal{B}_n(x) := \mathbb{E}[(\hat{\theta}_S(x) - \theta^*)^2] = v_n + b^2(x) + o(v_n) + o(b^2(x)). \quad (10)$$

Proof. Conditionally on $\mathcal{F}_C$, $\hat{\theta}_S(x) - \theta^* = (\hat{\theta}_S(x) - \theta_x) + \Delta(x)$. The first term has conditional mean 0 and conditional variance $\sigma_S^2(x)/m_n + o(m_n^{-1})$ by Assumption 1.2; the second term is $\mathcal{F}_C$ -measurable. Hence

$$\mathbb{E}[(\hat{\theta}_S - \theta^*)^2 \mid \mathcal{F}_C] = \frac{\sigma_S^2(x)}{m_n} + \Delta(x)^2 + o(m_n^{-1}).$$

Taking unconditional expectations and using $\mathbb{E}[\sigma_S^2(x)/m_n] = v_n\{1 + o(1)\}$ together with $\mathbb{E}[\Delta(x)^2] = b^2(x)$ yields the claim. $\square$

1.6 Working model

Assumption 1.3 (Power-law calibration). There exist $c > 0$, $\beta > 0$, and a deterministic sequence $\eta_x \rightarrow 0$ as $x \rightarrow \infty$ such that

$$b^2(x) = c x^{-2\beta} (1 + \eta_x) \quad \text{for all } x \geq 1. \quad (11)$$

Assumption 1.4 (Synthetic-variance regime). There exist $v_0 > 0$, $\rho \geq 0$, and $\zeta_n \rightarrow 0$ such that

$$v_n = v_0 n^{-\rho} (1 + \zeta_n). \quad (12)$$

The case $m_n \asymp n^\rho$ with constant per-sample variance gives Assumption 1.4. The headline scenario $m_n \asymp n$ corresponds to $\rho = 1$.

Remark 1.2 (Plateau bias and misspecification). The plateau-bias generalization $b^2(x) = b_\infty^2 + c x^{-2\beta} (1 + \eta_x)$ with $b_\infty^2 \geq 0$ relaxes the power-law assumption to allow irreducible bias from generator misspecification. Setting $b_\infty^2 = 0$ recovers Assumption 1.3. The plateau case is treated separately in Proposition 2.12.

Remark 1.3 (Vector estimands). For $\theta^* \in \mathbb{R}^d$, replace a by a positive-definite covariance matrix Σ_R and $b^2(x)$ by the matrix $\mathbb{E}[\Delta(x)\Delta(x)^\top]$; replace scalar variances by traces of the corresponding matrices. The matrix-weighted optimum is recorded in Remark 2.3 below.

1.7 Two estimator classes

We study two estimator classes built from the calibrated generator and the sample-split partition.

Linear allocation estimator. For the linear configuration ($V_x = \emptyset$) and a mixing weight $\alpha \in [0, 1]$, define

$$\hat{\theta}^L(\alpha, x) := \alpha \hat{\theta}_R(E_x) + (1 - \alpha) \hat{\theta}_S(x). \quad (13)$$

Validation-debiased (PPI) estimator. Assume the synthetic distribution admits a measurable functional representative $\tilde{g}_x : \mathcal{X} \rightarrow \mathbb{R}$ satisfying

$$\mathbb{E}_{X \sim P^*} [\tilde{g}_x(X) \mid \mathcal{F}_C] = \theta_x. \quad (14)$$

For instance, when $\theta(P) = \mathbb{E}_P[g(W)]$ with $W = (X, Y)$ and P_x has the same X -marginal as P^* , take $\tilde{g}_x(X) = \mathbb{E}_{P_x}[g(W) \mid X]$. For the PPI configuration ($E_x = \emptyset$, $V_x = [n] \setminus C_x$), define

$$\hat{\theta}^{\text{PP}}(x) := \hat{\theta}_S(x) - \hat{\theta}_S^V(x) + \hat{\theta}_R^V, \quad (15)$$

with

$$\hat{\theta}_S^V(x) := \frac{1}{n_v} \sum_{i \in V_x} \tilde{g}_x(X_i), \quad \hat{\theta}_R^V := \frac{1}{n_v} \sum_{i \in V_x} g(X_i), \quad n_v := n - x.$$

For mean estimands $\theta^* = \mathbb{E}_{P^*}[g(W)]$, $\hat{\theta}_R^V = n_v^{-1} \sum_{i \in V_x} g(W_i)$.

Lemma 1.4 (Conditional independence). *Under Assumptions 1.1 and 1.2 and the sample-splitting protocol of Section 1.2:*

- (i) *In the linear configuration, $\hat{\theta}_R(E_x) \perp\!\!\!\perp (\mathcal{F}_C, \{Z_j\}_{j=1}^{m_n})$. In particular, $\hat{\theta}_R(E_x) \perp\!\!\!\perp \hat{\theta}_S(x)$ unconditionally.*
- (ii) *In the PPI configuration, $(\{X_i\}_{i \in V_x}) \perp\!\!\!\perp \mathcal{F}_C$ and, conditionally on $\mathcal{F}_C$, $\hat{\theta}_S(x) \perp\!\!\!\perp (\hat{\theta}_S^V(x), \hat{\theta}_R^V)$.*

Proof. Both statements follow from the i.i.d. structure of $\{X_i\}$, the disjointness of C_x, E_x, V_x , and the independence of $\{U_j\}$ (and hence of the synthetic samples $\{Z_j\}$) from $\{X_i\}$. $\square$

2 Theory I: Linear Allocation Estimator

Throughout this section we work with the linear configuration: $E_x = [n] \setminus C_x$, $V_x = \emptyset$, and $\hat{\theta}^L(\alpha, x)$ defined by Eq. (13). Write $V_R(x) = a/(n-x)$ and $\mathcal{B}_n(x) = v_n + b^2(x)$.

2.1 MSE decomposition and optimal mixing

Lemma 2.1 (Working MSE). *Under Assumptions 1.1 and 1.2,*

$$\mathbb{E}[(\hat{\theta}^L(\alpha, x) - \theta^*)^2] = \alpha^2 V_R(x) + (1 - \alpha)^2 \mathcal{B}_n(x) + \mathcal{R}_n(\alpha, x), \quad (16)$$

where the remainder $\mathcal{R}_n(\alpha, x) = o(V_R(x)) + o(\mathcal{B}_n(x))$ uniformly in $\alpha \in [0, 1]$.

Proof. Write $\xi_R = \hat{\theta}_R(E_x) - \theta^*$ and $\xi_S = \hat{\theta}_S(x) - \theta^*$. By Lemma 1.4(i), $\xi_R \perp \xi_S$, hence $\mathbb{E}[(\alpha \xi_R + (1 - \alpha) \xi_S)^2] = \alpha^2 \mathbb{E}[\xi_R^2] + (1 - \alpha)^2 \mathbb{E}[\xi_S^2] + 2\alpha(1 - \alpha) \mathbb{E}[\xi_R] \mathbb{E}[\xi_S]$. By Assumption 1.1, $\mathbb{E}[\xi_R] = \mathbb{E}[r_R(E_x)] = 0$, so the cross term vanishes exactly. The first two terms equal $\alpha^2 V_R(x) + (1 - \alpha)^2 \mathcal{B}_n(x)$ up to the stated remainders by Assumption 1.1 and Lemma 1.1. $\square$

We work hereafter with the leading-order MSE

$$\overline{\text{MSE}}_n(\alpha, x) := \alpha^2 V_R(x) + (1 - \alpha)^2 \mathcal{B}_n(x). \quad (17)$$

The actual MSE differs by a $1 + o(1)$ multiplicative factor at the optimum, which is sufficient for relative-oracle results.

Proposition 2.2 (Optimal mixing weight and profiled risk). *For every $x \in [0, n)$ with $\mathcal{B}_n(x) < \infty$, the unique minimizer of $\overline{\text{MSE}}_n(\cdot, x)$ on $[0, 1]$ is*

$$\alpha^*(x) = \frac{\mathcal{B}_n(x)}{V_R(x) + \mathcal{B}_n(x)} \in (0, 1), \quad (18)$$

and the profiled risk is

$$R_n(x) := \min_{\alpha \in [0, 1]} \overline{\text{MSE}}_n(\alpha, x) = \frac{V_R(x) \mathcal{B}_n(x)}{V_R(x) + \mathcal{B}_n(x)} = \frac{a \mathcal{B}_n(x)}{a + (n - x) \mathcal{B}_n(x)}. \quad (19)$$

Proof. $\overline{\text{MSE}}_n(\alpha, x)$ is strictly convex in α with second derivative $2[V_R(x) + \mathcal{B}_n(x)] > 0$. Setting the first derivative $2\alpha V_R(x) - 2(1 - \alpha)\mathcal{B}_n(x)$ to zero yields Eq. (18). Both $V_R(x), \mathcal{B}_n(x) > 0$, so the minimizer is interior. Substituting Eq. (18) into Eq. (17) gives the harmonic-risk formula Eq. (19); the second equality uses $V_R(x) = a/(n - x)$. $\square$

Remark 2.3 (Vector mixing). For $\theta^* \in \mathbb{R}^d$, the trace-MSE-minimizing matrix weight $W^*(x)$ on $\hat{\theta}_R$ within the symmetric positive-definite class satisfies $W^*(x) = \Sigma_B(x)(\Sigma_R(x) + \Sigma_B(x))^{-1}$ when $\Sigma_R(x)$ and $\Sigma_B(x)$ commute, where $\Sigma_R(x) = \Sigma_R/(n - x)$ and $\Sigma_B(x) = \Sigma_S/m_n + \mathbb{E}[\Delta(x)\Delta(x)^\top]$. In the noncommuting case, the trace-optimal symmetric weight is the unique positive-definite solution of $W\Sigma_R(x) + \Sigma_R(x)W = 2W^{1/2}\Sigma_B(x)W^{1/2}$, obtained by spectral decomposition; the resulting profiled risk admits the same harmonic-mean structure in the GLS norm. We work in the scalar case throughout; componentwise application yields a valid, possibly suboptimal vector estimator.

2.2 Critical-point structure of the profiled risk

The next lemma is the technical workhorse: it determines the global shape of $R_n(\cdot)$ on $(0, n)$. It establishes that for $\beta > 1/2$ the first-order condition has *two* interior roots in the asymptotic regime of interest, and that only the larger root corresponds to a local minimum. The earlier

draft of this paper implicitly selected the larger root; the analysis below makes the selection rigorous and identifies the smaller root as a local maximum.

Define

$$\Phi(x) := \mathcal{B}_n(x)^2 x^{2\beta+1}, \quad x > 0, \quad (20)$$

and

$$H(x) := a \mathcal{B}'_n(x) + \mathcal{B}_n(x)^2. \quad (21)$$

Under Assumption 1.3, $\mathcal{B}'_n(x) = -2\beta cx^{-(2\beta+1)} + o(x^{-(2\beta+1)})$ where the $o(\cdot)$ comes from differentiating the $1 + \eta_x$ factor; since the perturbation analysis is identical to the leading-order case, we treat the $1 + \eta_x$ factor as 1 within the lemmas, and record the perturbation argument in Remark 2.8.

Lemma 2.4 (Gradient of the profiled risk). *For every $x \in (0, n)$,*

$$R'_n(x) = \frac{a H(x)}{[a + (n-x) \mathcal{B}_n(x)]^2} = \frac{a}{[a + (n-x) \mathcal{B}_n(x)]^2} \cdot \frac{\Phi(x) - 2\beta ac}{x^{2\beta+1}}. \quad (22)$$

In particular, $\text{sign}(R'_n(x)) = \text{sign}(\Phi(x) - 2\beta ac)$.

Proof. Write $R_n = u/w$ with $u(x) = a\mathcal{B}_n(x)$ and $w(x) = a + (n-x)\mathcal{B}_n(x)$. Then $u'(x) = a\mathcal{B}'_n(x)$ and $w'(x) = -\mathcal{B}_n(x) + (n-x)\mathcal{B}'_n(x)$. Compute:

$$\begin{aligned} u'w - uw' &= a\mathcal{B}'_n[a + (n-x)\mathcal{B}_n] - a\mathcal{B}_n[-\mathcal{B}_n + (n-x)\mathcal{B}'_n] \\ &= a^2\mathcal{B}'_n + a(n-x)\mathcal{B}_n\mathcal{B}'_n + a\mathcal{B}_n^2 - a(n-x)\mathcal{B}_n\mathcal{B}'_n \\ &= a^2\mathcal{B}'_n + a\mathcal{B}_n^2 = aH. \end{aligned}$$

The cancellation of the cross terms $\pm a(n-x)\mathcal{B}_n\mathcal{B}'_n$ is the algebraic reason the optimum does not depend on $n-x$ at first order. Multiplying H by $x^{2\beta+1}$ gives $H \cdot x^{2\beta+1} = \mathcal{B}_n^2 x^{2\beta+1} - 2\beta ac = \Phi(x) - 2\beta ac$. $\square$

Lemma 2.5 (Shape of Φ). *Under Assumption 1.3, the function $\Phi(x) = (v_n + cx^{-2\beta})^2 x^{2\beta+1}$ on $(0, \infty)$ satisfies:*

- (a) $\Phi'(x) = x^{2\beta} \mathcal{B}_n(x) [(2\beta+1)v_n + (1-2\beta)cx^{-2\beta}]$.
- (b) *If $\beta \leq 1/2$, then $\Phi'(x) > 0$ for all $x > 0$ (with $v_n > 0$): Φ is strictly increasing, with $\Phi(0^+) = c^2 \mathbf{1}\{\beta = 1/2\}$ and $\Phi(\infty) = +\infty$.*
- (c) *If $\beta > 1/2$, then Φ is strictly U-shaped on $(0, \infty)$ with unique minimum at*

$$x^\dagger = \left(\frac{c(2\beta-1)}{(2\beta+1)v_n} \right)^{1/(2\beta)}, \quad (23)$$

and minimum value

$$\Phi^\dagger := \Phi(x^\dagger) = K(\beta) c^{(2\beta+1)/(2\beta)} v_n^{(2\beta-1)/(2\beta)}, \quad K(\beta) := \frac{16\beta^2}{(2\beta+1)^{(2\beta+1)/(2\beta)} (2\beta-1)^{(2\beta-1)/(2\beta)}}. \quad (24)$$

$$\Phi(0^+) = \Phi(\infty) = +\infty.$$

Proof. (a) Direct differentiation: $\Phi'(x) = 2\mathcal{B}_n(x)\mathcal{B}'_n(x)x^{2\beta+1} + \mathcal{B}_n(x)^2(2\beta+1)x^{2\beta} = x^{2\beta}\mathcal{B}_n(x)[2\mathcal{B}'_n(x)x + (2\beta+1)\mathcal{B}_n(x)]$. Substituting $\mathcal{B}'_n(x) = -2\beta cx^{-(2\beta+1)}$ and $\mathcal{B}_n(x) = v_n + cx^{-2\beta}$: $2\mathcal{B}'_n(x)x + (2\beta+1)\mathcal{B}_n(x) = -4\beta cx^{-2\beta} + (2\beta+1)v_n + (2\beta+1)cx^{-2\beta} = (2\beta+1)v_n + (1-2\beta)cx^{-2\beta}$.

(b) For $\beta \leq 1/2$, $1-2\beta \geq 0$ and $v_n > 0$, so the bracketed term in (a) is strictly positive on $(0, \infty)$; hence $\Phi' > 0$ and Φ is strictly increasing. As $x \rightarrow 0^+$, $\mathcal{B}_n(x) \sim cx^{-2\beta}$ so $\Phi(x) \sim c^2 x^{1-2\beta}$.

For $\beta < 1/2$ this tends to 0; for $\beta = 1/2$, $\Phi(x) = (v_n x + c)^2 \rightarrow c^2$. As $x \rightarrow \infty$, $\mathcal{B}_n(x) \rightarrow v_n$ so $\Phi(x) \sim v_n^2 x^{2\beta+1} \rightarrow \infty$ (using $v_n > 0$).

(c) For $\beta > 1/2$, $1 - 2\beta < 0$. The bracket $(2\beta + 1)v_n + (1 - 2\beta)cx^{-2\beta}$ is strictly decreasing in $cx^{-2\beta}$, equals 0 when $cx^{-2\beta} = (2\beta + 1)v_n/(2\beta - 1)$, equivalently at $x = x^\dagger$ given by Eq. (23). For $x < x^\dagger$, $cx^{-2\beta} > (2\beta + 1)v_n/(2\beta - 1)$, so the bracket is negative and $\Phi' < 0$. For $x > x^\dagger$, the bracket is positive, $\Phi' > 0$. Hence Φ is strictly decreasing on $(0, x^\dagger)$ and strictly increasing on $(x^\dagger, \infty)$, with unique minimum at $x^\dagger$.

To compute $\Phi^\dagger = \Phi(x^\dagger)$, use $cx^{\dagger-2\beta} = (2\beta + 1)v_n/(2\beta - 1)$, so

$$\mathcal{B}_n(x^\dagger) = v_n + \frac{(2\beta + 1)v_n}{2\beta - 1} = v_n \cdot \frac{4\beta}{2\beta - 1}.$$

From Eq. (23), $x^{\dagger 2\beta+1} = [c(2\beta - 1)/((2\beta + 1)v_n)]^{(2\beta+1)/(2\beta)}$. Therefore

$$\begin{aligned} \Phi^\dagger &= \left(\frac{4\beta v_n}{2\beta - 1}\right)^2 \cdot \left(\frac{c(2\beta - 1)}{(2\beta + 1)v_n}\right)^{(2\beta+1)/(2\beta)} \\ &= \frac{16\beta^2 c^{(2\beta+1)/(2\beta)}}{(2\beta + 1)^{(2\beta+1)/(2\beta)}(2\beta - 1)^{(2\beta-1)/(2\beta)}} \cdot v_n^{(2\beta-1)/(2\beta)}, \end{aligned}$$

where we used $(2\beta + 1)/(2\beta) - 2 = (1 - 2\beta)/(2\beta)$ to combine the $(2\beta - 1)$ powers, and $2 - (2\beta + 1)/(2\beta) = (2\beta - 1)/(2\beta)$ for the v_n exponent. The endpoint behavior follows from the same expansions used in (b): as $x \rightarrow 0^+$, $\Phi \sim c^2 x^{1-2\beta} \rightarrow +\infty$ since $1 - 2\beta < 0$; as $x \rightarrow \infty$, $\Phi \sim v_n^2 x^{2\beta+1} \rightarrow +\infty$. $\square$

Lemma 2.6 (Critical points of R_n). *Assume Assumptions 1.3 and 1.4 with $a, c, v_n > 0$.*

(i) *If $\beta \leq 1/2$: H has at most one zero on $(0, \infty)$. When it exists, it is the unique interior critical point of R_n and is a strict local minimum. For $\beta < 1/2$ the zero exists for every $v_n > 0$. For $\beta = 1/2$ the zero exists if and only if $a > c$.*

(ii) *If $\beta > 1/2$: Set $\Phi^\dagger$ as in Eq. (24).*

(a) *If $\Phi^\dagger > 2\beta ac$: R_n is strictly increasing on $(0, n)$; no interior critical point.*

(b) *If $\Phi^\dagger = 2\beta ac$: R_n has a unique inflection point at $x^\dagger$ but no strict local extremum.*

(c) *If $\Phi^\dagger < 2\beta ac$: H has exactly two zeros $x_1 < x^\dagger < x_2$ in $(0, \infty)$. The point x_1 is a strict local maximum of R_n , and x_2 is a strict local minimum of R_n .*

Under Assumption 1.4 with $\rho > 0$, $\Phi^\dagger = K(\beta)c^{(2\beta+1)/(2\beta)}v_0^{(2\beta-1)/(2\beta)}n^{-\rho(2\beta-1)/(2\beta)} \rightarrow 0$, so case (iic) holds for all sufficiently large n .

Proof. By Lemma 2.4, the zeros of H on $(0, n)$ are exactly the points where $\Phi = 2\beta ac$, and the sign of R'_n matches the sign of $\Phi - 2\beta ac$.

(i) $\beta \leq 1/2$. By Lemma 2.5(b), Φ is strictly increasing. Hence $\{x : \Phi(x) = 2\beta ac\}$ contains at most one element. When it exists (call it x°), $\Phi - 2\beta ac < 0$ on $(0, x^\circ)$ and > 0 on (x°, ∞) , so $R'_n < 0$ before x° and $R'_n > 0$ after; thus x° is a strict local minimum. For $\beta < 1/2$, $\Phi(0^+) = 0 < 2\beta ac < \Phi(\infty) = \infty$ so x° exists by continuity. For $\beta = 1/2$, $\Phi(0^+) = c^2$ and $\Phi(\infty) = \infty$, so x° exists iff $c^2 < 2\beta ac = ac$, i.e. $c < a$.

(ii) $\beta > 1/2$. By Lemma 2.5(c), Φ is U-shaped with minimum $\Phi^\dagger$.

Case (a). If $\Phi^\dagger > 2\beta ac$, then $\Phi(x) > \Phi^\dagger > 2\beta ac$ for all $x > 0$, so $R'_n > 0$ throughout $(0, n)$.

Case (b). If $\Phi^\dagger = 2\beta ac$, $\Phi - 2\beta ac \geq 0$ with equality only at $x^\dagger$; thus $R'_n \geq 0$ with $R'_n(x^\dagger) = 0$, but R'_n does not change sign, so $x^\dagger$ is an inflection.

Case (c). If $\Phi^\dagger < 2\beta ac$: Since $\Phi(0^+) = \Phi(\infty) = +\infty$ and Φ is continuous and strictly U-shaped with minimum $\Phi^\dagger < 2\beta ac$, the equation $\Phi(x) = 2\beta ac$ has exactly two solutions $x_1 \in (0, x^\dagger)$ and $x_2 \in (x^\dagger, \infty)$ by the intermediate value theorem applied to each strictly

monotone branch. On $(0, x_1)$, $\Phi > 2\beta ac$ so $R'_n > 0$; on (x_1, x_2) , $\Phi < 2\beta ac$ so $R'_n < 0$; on (x_2, ∞) , $\Phi > 2\beta ac$ so $R'_n > 0$. Hence x_1 is a strict local maximum and x_2 a strict local minimum of R_n .

The final claim follows from Eq. (24) and Assumption 1.4: with $v_n = v_0 n^{-\rho}(1 + o(1))$, $\Phi^\dagger \asymp n^{-\rho(2\beta-1)/(2\beta)} \rightarrow 0$ when $\rho > 0$ and $\beta > 1/2$, so $\Phi^\dagger < 2\beta ac$ for n large. $\square$

Remark 2.7 (Selection of the active root). In case (iic), the global minimizer of R_n on $[0, n]$ is one of $\{0, x_2, n\}$: the local maximum x_1 is always dominated by the boundary 0. The active interior solution—i.e., the only interior local minimum—is therefore x_2 . We refer to this as the *active interior root* and write $x_n^\circ := x_2$.

Remark 2.8 (Perturbative argument for the $1 + \eta_x$ factor). Throughout this section we suppress the $1 + \eta_x$ factor in $b^2(x) = cx^{-2\beta}(1 + \eta_x)$. The claims of Lemmas 2.5 and 2.6 go through unchanged for the perturbed model: the bracket in $\Phi'(x)$ becomes $(2\beta + 1)v_n + (1 - 2\beta)cx^{-2\beta} + o(cx^{-2\beta})$, which has the same sign behavior as the unperturbed bracket on $(0, \infty)$. The location of $x^\dagger$ shifts by a $1 + o(1)$ factor, and the value $\Phi^\dagger$ by the same. The phase-diagram exponents in Theorem 2.11 below are unaffected.

2.3 Exact first-order condition and safe-improvement

Proposition 2.9 (Exact first-order condition). *Under Assumptions 1.3 and 1.4 with $v_n > 0$, every interior critical point $x \in (0, n)$ of R_n satisfies*

$$\boxed{(v_n + cx^{-2\beta})^2 = 2\beta ac x^{-(2\beta+1)}}, \quad (25)$$

or equivalently, after multiplication by $x^{2\beta+1}$,

$$v_n^2 x^{2\beta+1} + 2cv_n x + c^2 x^{1-2\beta} = 2\beta ac. \quad (26)$$

For $\beta \leq 1/2$ this equation has at most one solution; for $\beta > 1/2$ it has at most two, and the active interior root is the larger.

Proof. By Lemmas 2.4 and 2.6, interior critical points are exactly the solutions of $H(x) = 0$, equivalently $\Phi(x) = 2\beta ac$. Writing $\Phi(x) = (v_n + cx^{-2\beta})^2 x^{2\beta+1}$ and dividing by $x^{2\beta+1}$ gives Eq. (25); expanding the square gives Eq. (26). Multiplicity follows from Lemma 2.6. $\square$

Proposition 2.10 (Exact safe-improvement condition). *For every $x \in (0, n)$,*

$$R_n(x) < \frac{a}{n} \iff \boxed{x \mathcal{B}_n(x) < a}. \quad (27)$$

Equivalently, $x(v_n + cx^{-2\beta}) < a$.

Proof. $R_n(x) = a\mathcal{B}_n(x)/[a + (n - x)\mathcal{B}_n(x)] < a/n$ iff $n\mathcal{B}_n < a + (n - x)\mathcal{B}_n$ iff $x\mathcal{B}_n < a$. The boundary risk is $R_n(0) = a/n$. $\square$

2.4 Phase diagram

Theorem 2.11 (Allocation phase diagram). *Assume Assumptions 1.3 and 1.4. Let x_n° denote the active interior root from Remark 2.7 (when it exists) and $x_n^* \in \arg \min_{x \in [0, n]} R_n(x)$ the global oracle. As $n \rightarrow \infty$:*

Regime I (*persistent synthetic variance*, $\rho = 0$). *Every interior critical point is $O(1)$. Hence $x_n^*/n \rightarrow 0$ regardless of β .*

Regime II (*slow learning*, $\beta < 1/2$, $\rho > 0$). *$x_n^\circ \rightarrow x_\infty := (2\beta a/c)^{1/(1-2\beta)} \in (0, \infty)$, hence $x_n^\circ = O(1)$ and $x_n^*/n \rightarrow 0$. The first-order rate of $R_n(x_n^*)$ matches the all-real rate a/n , with strict finite-sample shrinkage gain*

$$R_n(x_n^*) = \frac{a}{n} - \frac{a^2/n^2}{cx_\infty^{-2\beta}} + o(n^{-2}).$$

Regime III (sublinear-calibration, $\beta > 1/2$, $0 < \rho < \beta + \frac{1}{2}$). For all sufficiently large n , the active interior root exists and satisfies

$$x_n^\circ = \left(\frac{2\beta ac}{v_0^2} \right)^{1/(2\beta+1)} n^{2\rho/(2\beta+1)} \{1 + o(1)\}. \quad (28)$$

Consequently $\lambda_n^\circ = x_n^\circ/n \asymp n^{2\rho/(2\beta+1)-1} \rightarrow 0$, and the safe-improvement condition holds asymptotically: $x_n^\circ \mathcal{B}_n(x_n^\circ) \rightarrow 0 < a$. Hence $x_n^* = x_n^\circ \{1 + o(1)\}$ and

$$R_n(x_n^*) = \frac{V_R(x_n^\circ) v_n}{V_R(x_n^\circ) + v_n} \{1 + o(1)\} = \begin{cases} \frac{a}{n} \{1 + o(1)\} & \text{if } \rho < 1, \\ \frac{av_0}{(a + v_0)n} \{1 + o(1)\} & \text{if } \rho = 1, \\ \frac{v_0}{n^\rho} \{1 + o(1)\} & \text{if } \rho > 1. \end{cases} \quad (29)$$

Regime IV (full calibration, $\beta > 1/2$, $\rho > \beta + \frac{1}{2}$). The asymptotic interior root from Eq. (28) exceeds n for n large; the boundary $x = n$ is feasible and $\mathcal{B}_n(n) = v_n + cn^{-2\beta} = o(n^{-1})$ since both $\rho > 1$ and $2\beta > 1$. Thus $\lambda_n^* \rightarrow 1$ and $R_n(x_n^*) = v_n + cn^{-2\beta} = o(n^{-1})$, beating the all-real rate.

Regime V (parametric knife-edge, $\beta = 1/2$). An interior root exists iff $a > c$; in that case

$$x_n^\circ = \frac{\sqrt{ac} - c}{v_0} n^\rho \{1 + o(1)\}.$$

The boundary regime depends on ρ and constants:

- $\rho < 1$: $\lambda_n^\circ \rightarrow 0$, $R_n(x_n^*) = a/n \{1 + o(1)\}$ with strict finite-sample improvement.
- $\rho = 1$: $\lambda_n^\circ \rightarrow (\sqrt{ac} - c)/v_0 \in (0, 1)$ provided $\sqrt{ac} - c < v_0$, with $R_n(x_n^*) \asymp n^{-1}$.
- $\rho > 1$: x_n° exceeds n ; the global oracle is at the boundary if $c < a$, with $\lambda_n^* \rightarrow 1$ and $R_n(x_n^*) = v_n + c/n = O(\max(n^{-\rho}, n^{-1}))$.

If $a \leq c$, no interior root exists, the safe-improvement condition fails asymptotically, and $x_n^* = 0$ is the global oracle.

Proof. We analyze Eq. (26) in each regime, using Lemma 2.6 for existence and identification of the active root.

Regime I ($\rho = 0$, $v_n \rightarrow v_0 > 0$). Equation (26) reads $v_0^2 x^{2\beta+1} + 2cv_0 x + c^2 x^{1-2\beta} = 2\beta ac + o(1)$, with no n -dependence in the leading coefficients. By Lemma 2.6, any active root is determined by the constants (a, c, v_0, β) alone and is therefore $O(1)$. Compared to the unbounded n , $x_n^\circ/n \rightarrow 0$. The candidates $\{0, x_n^\circ, n\}$ are evaluated by direct substitution; the global minimizer is one of $\{0, x_n^\circ\}$ since $R_n(n) = v_n + cn^{-2\beta} \rightarrow v_0$ does not vanish.

Regime II ($\beta < 1/2$, $\rho > 0$). With $v_n \rightarrow 0$, Eq. (26) reduces to $c^2 x^{1-2\beta} = 2\beta ac + o(1)$ at any solution where x remains $O(1)$. Solving: $x^{1-2\beta} = 2\beta a/c + o(1)$, so $x_n^\circ \rightarrow x_\infty = (2\beta a/c)^{1/(1-2\beta)}$. (The terms $v_n^2 x^{2\beta+1} = O(v_n^2)$ and $2cv_n x = O(v_n)$ are $o(1)$ relative to the constant RHS for fixed x .) Conversely, suppose for contradiction $x_n^\circ \rightarrow \infty$. Then $c^2 x^{1-2\beta} \rightarrow \infty$ since $1 - 2\beta > 0$, contradicting Eq. (26). So $x_n^\circ = \Theta(1)$ as claimed.

For the rate: $V_R(x_n^\circ) = a/(n - x_n^\circ) \sim a/n$ and $\mathcal{B}_n(x_n^\circ) = cx_n^{\circ-2\beta} + o(1) \rightarrow cx_\infty^{-2\beta} > 0$. Substituting in Eq. (19):

$$R_n(x_n^\circ) = \frac{V_R \mathcal{B}_n}{V_R + \mathcal{B}_n} = V_R \cdot \frac{1}{1 + V_R/\mathcal{B}_n} = V_R - \frac{V_R^2}{\mathcal{B}_n} + O((V_R/\mathcal{B}_n)^3),$$

which yields $R_n(x_n^\circ) = a/n - (a/n)^2/(cx_\infty^{-2\beta}) + o(n^{-2})$. Safe-improvement: $x_\infty \mathcal{B}_n(x_\infty) \rightarrow x_\infty \cdot cx_\infty^{-2\beta} = cx_\infty^{1-2\beta} = c(2\beta a/c) = 2\beta a < a$ since $\beta < 1/2$. Hence $R_n(x_n^\circ) < a/n$ asymptotically, so $x_n^* = x_n^\circ \{1 + o(1)\}$.

Regime III ($\beta > 1/2$, $0 < \rho < \beta + \frac{1}{2}$). By Lemma 2.6(iic), the active root x_n° exists for n large. Conjecture $cx_n^{\circ-2\beta} \ll v_n$ at the root, so $\mathcal{B}_n(x_n^\circ) = v_n\{1 + o(1)\}$. Equation (25) then gives $v_n^2 = 2\beta ac x_n^{\circ-(2\beta+1)}\{1 + o(1)\}$, equivalently

$$x_n^\circ = \left(\frac{2\beta ac}{v_n^2}\right)^{1/(2\beta+1)}\{1 + o(1)\} = \left(\frac{2\beta ac}{v_0^2}\right)^{1/(2\beta+1)} n^{2\rho/(2\beta+1)}\{1 + o(1)\}.$$

We verify the conjecture: $cx_n^{\circ-2\beta} = c \cdot [2\beta ac/v_0^2]^{-2\beta/(2\beta+1)} n^{-4\beta\rho/(2\beta+1)}$, while $v_n \asymp n^{-\rho}$. The ratio is $\Theta(n^{-4\beta\rho/(2\beta+1)+\rho}) = \Theta(n^{-\rho(2\beta-1)/(2\beta+1)})$, which tends to 0 since $\beta > 1/2$ and $\rho > 0$. Hence $cx_n^{\circ-2\beta} = o(v_n)$, confirming the conjecture.

Feasibility: $x_n^\circ/n \asymp n^{2\rho/(2\beta+1)-1}$, with exponent $2\rho/(2\beta+1) - 1 < 0$ iff $\rho < \beta + \frac{1}{2}$. So $x_n^\circ = o(n)$. Safe-improvement: $x_n^\circ \mathcal{B}_n(x_n^\circ) \asymp n^{2\rho/(2\beta+1)} \cdot n^{-\rho} = n^{-\rho(2\beta-1)/(2\beta+1)} \rightarrow 0 < a$. Hence $R_n(x_n^\circ) < a/n$ and $R_n(x_n^\circ) < R_n(n)$ (the latter shown in Regime IV), so $x_n^* = x_n^\circ$.

For the risk asymptotics: $V_R(x_n^\circ) = a/(n - x_n^\circ) = a/n\{1 + o(1)\}$ since $x_n^\circ = o(n)$. $\mathcal{B}_n(x_n^\circ) = v_n\{1 + o(1)\}$. The harmonic-mean formula Eq. (19) gives the three subcases by direct substitution.

Regime IV ($\beta > 1/2$, $\rho > \beta + \frac{1}{2}$). The asymptotic Regime-III formula gives $x_n^\circ \asymp n^{2\rho/(2\beta+1)}$ with exponent > 1 , so $x_n^\circ > n$ eventually—*infeasible*. Compare $R_n(0) = a/n$ to $R_n(n) = v_n + cn^{-2\beta}$. With $\rho > \beta + \frac{1}{2} > 1$ and $2\beta > 1$: $v_n = O(n^{-\rho}) = o(n^{-1})$ and $cn^{-2\beta} = o(n^{-1})$, so $R_n(n) = o(n^{-1}) < a/n$. Hence $x_n^* = n\{1 + o(1)\}$ and $\lambda_n^* \rightarrow 1$.

Regime V ($\beta = 1/2$). Equation (26) becomes $(v_n x + c)^2 = ac$ after multiplication by x^2 and taking the positive square root: $v_n x + c = \sqrt{ac}$ (the positive root; the negative root gives $x < 0$). Solving: $x = (\sqrt{ac} - c)/v_n$, which is positive iff $a > c$. The asymptotic order follows from $v_n = v_0 n^{-\rho}\{1 + o(1)\}$.

If $a > c$ and $\rho > 1$, $x_n^\circ \sim ((\sqrt{ac} - c)/v_0)n^\rho$ exceeds n for n large; compare boundaries. $R_n(n) = v_n + c/n \sim c/n$ since $\rho > 1$. This beats a/n iff $c < a$, which holds by hypothesis. Hence $\lambda_n^* \rightarrow 1$.

If $a \leq c$: $\sqrt{ac} \leq c$ so the formula yields $x_n^\circ \leq 0$; no interior root. Safe-improvement at any $x > 0$ requires $x(v_n + c/x) < a$, i.e., $v_n x + c < a$, which fails since $c \geq a$. So $x_n^* = 0$. $\square$

2.5 Robustness to plateau bias

The following result addresses the plateau-bias generalization of Remark 1.2, in which the synthetic generator is misspecified so that some bias is irreducible. This is the practically important case for foundation-model-generated synthetic data, where alignment-style misspecification need not vanish with calibration size.

Proposition 2.12 (Plateau-bias regime). *Replace Assumption 1.3 by $b^2(x) = b_\infty^2 + cx^{-2\beta}\{1 + o(1)\}$ with $b_\infty^2 > 0$ and $\beta > 0$. Suppose Assumption 1.4 holds with $\rho \geq 0$. Then for n sufficiently large, the active interior root x_n° exists and satisfies*

$$x_n^\circ \leq \left(\frac{2\beta ac}{b_\infty^4}\right)^{1/(2\beta+1)} = O(1). \quad (30)$$

Consequently $\lambda_n^\circ \rightarrow 0$ for any $\beta > 0$ and any $\rho \geq 0$. The optimal risk satisfies $R_n(x_n^) = a/n\{1 + o(1)\}$: synthetic augmentation provides at most a finite-sample shrinkage gain and no first-order rate improvement. The shrinkage gain is positive iff $b_\infty^2 < a/x_n^\circ$.*

Proof. Now $\mathcal{B}_n(x) = v_n + b_\infty^2 + cx^{-2\beta} \geq b_\infty^2 > 0$ for all $x > 0$. The first-order condition $\mathcal{B}_n^2 = 2\beta acx^{-(2\beta+1)}$ requires $b_\infty^4 \leq \mathcal{B}_n^2 = 2\beta acx^{-(2\beta+1)}$, hence $x^{2\beta+1} \leq 2\beta ac/b_\infty^4$, giving Eq. (30). Hence $x_n^\circ = O(1)$.

For the risk: $V_R(x_n^\circ) \sim a/n$ and $\mathcal{B}_n(x_n^\circ) \geq b_\infty^2 > 0$ is bounded away from zero. Hence $V_R \ll \mathcal{B}_n$ asymptotically, and as in Regime II, $R_n(x_n^\circ) = V_R - V_R^2/\mathcal{B}_n + o(V_R^2/\mathcal{B}_n) = a/n - (a/n)^2/\mathcal{B}_n(x_n^\circ) + o(n^{-2})$. The shrinkage is positive iff $R_n(x_n^\circ) < a/n$, equivalent by Proposition 2.10 to $x_n^\circ \mathcal{B}_n(x_n^\circ) < a$, which becomes $x_n^\circ b_\infty^2 < a$ at leading order. $\square$

Proposition 2.12 is the technical statement underlying a practical reading: *plateau bias caps calibration at a constant budget regardless of β, ρ , and there is no first-order rate improvement over the all-real estimator*. The earlier phase-diagram regimes (especially Regime III) presume $b_\infty^2 = 0$. When b_∞^2 is small but positive, Regime III's $\Theta(n^{2\rho/(2\beta+1)})$ scaling holds in the regime $b_\infty^2 \leq cx_n^{\circ-2\beta}$, i.e., until calibration would push the bias below the plateau; beyond that, calibration is wasted and Proposition 2.12 takes over.

2.6 Adaptive estimator and oracle inequality

The oracle allocation x_n^* depends on the unknown quantities (a, c, β, v_n) . The practical method estimates the risk curve $R_n(\cdot)$ on a grid and minimizes the estimate. We avoid plug-in estimators that require recovery of β at parametric rate (which is generically infeasible), instead using grid minimization that needs only uniform relative consistency of the risk estimator.

Algorithm 1 (Grid-based corrected Generalized Neyman allocation).

1. Choose a grid $\mathcal{G}_n \subset \{1, \dots, n-1\}$ and number of folds K .
2. Partition $[n]$ into K folds; for each fold, the within-fold estimator uses out-of-fold data for calibration-pilot and bias-curve estimation, and within-fold data for final estimation.
3. For each $x \in \mathcal{G}_n$:
 - (a) Estimate the influence-function variance $\hat{a}$ using the empirical variance of ψ evaluated on out-of-fold data.
 - (b) Calibrate the generator with x out-of-fold real observations, draw m_n synthetic observations, compute $\hat{\theta}_S(x)$, and estimate $\hat{v}_n$ from the synthetic sample.
 - (c) Estimate squared bias by sample-split moment matching: $\hat{b}^2(x) = [(\hat{\theta}_S(x) - \hat{\theta}_R^V)^2 - \widehat{\text{Var}}(\hat{\theta}_S(x)) - \widehat{\text{Var}}(\hat{\theta}_R^V)]_+$ on a held-out validation slice.
 - (d) Set $\hat{\mathcal{B}}_n(x) = \hat{v}_n + \hat{b}^2(x)$, $\hat{V}_R(x) = \hat{a}/(n-x)$, and $\hat{R}_n(x) = \hat{V}_R(x)\hat{\mathcal{B}}_n(x)/[\hat{V}_R(x) + \hat{\mathcal{B}}_n(x)]$.
4. Choose $\hat{x}_n = \arg \min_{x \in \mathcal{G}_n \cup \{0, n\}} \hat{R}_n(x)$, where $\hat{R}_n(0) = \hat{a}/n$ and $\hat{R}_n(n) = \hat{v}_n + \hat{b}^2(n)$.
5. Apply the safe fallback: if $\hat{x}_n \hat{\mathcal{B}}_n(\hat{x}_n) \geq \hat{a}$, set $\hat{x}_n = 0$.
6. Compute $\hat{\alpha} = \hat{\mathcal{B}}_n(\hat{x}_n)/[\hat{V}_R(\hat{x}_n) + \hat{\mathcal{B}}_n(\hat{x}_n)]$ and return $\hat{\theta} = \hat{\alpha}\hat{\theta}_R(\hat{x}_n) + (1 - \hat{\alpha})\hat{\theta}_S(\hat{x}_n)$.
7. Cross-fit by averaging across folds.

Theorem 2.13 (Adaptive oracle inequality). *Let $\mathcal{G}_n^+ := \mathcal{G}_n \cup \{0, n\}$, $x_n^* := \arg \min_{x \in \mathcal{G}_n^+} R_n(x)$, and $\hat{x}_n := \arg \min_{x \in \mathcal{G}_n^+} \hat{R}_n(x)$. Suppose the estimated risk is uniformly relatively consistent on the grid:*

$$\Delta_n := \sup_{x \in \mathcal{G}_n^+} \left| \frac{\hat{R}_n(x)}{R_n(x)} - 1 \right| \xrightarrow{p} 0. \quad (31)$$

Then

$$\frac{R_n(\hat{x}_n)}{R_n(x_n^*)} \xrightarrow{p} 1. \quad (32)$$

If furthermore the grid satisfies $\inf_{x \in \mathcal{G}_n} R_n(x)/\inf_{x \in [0, n]} R_n(x) \rightarrow 1$, then $R_n(\hat{x}_n)/\inf_{x \in [0, n]} R_n(x) \xrightarrow{p} 1$.

Proof. On the event $\{\Delta_n \leq \delta\}$, $(1 - \delta)R_n(x) \leq \widehat{R}_n(x) \leq (1 + \delta)R_n(x)$ for every $x \in \mathcal{G}_n^+$. By definition of $\hat{x}_n$, $\widehat{R}_n(\hat{x}_n) \leq \widehat{R}_n(x_n^*)$. Chaining:

$$(1 - \delta)R_n(\hat{x}_n) \leq \widehat{R}_n(\hat{x}_n) \leq \widehat{R}_n(x_n^*) \leq (1 + \delta)R_n(x_n^*).$$

Hence $R_n(\hat{x}_n)/R_n(x_n^*) \leq (1 + \delta)/(1 - \delta)$. Since $R_n(\hat{x}_n) \geq R_n(x_n^*)$ by oracle definition, the lower bound is 1. Setting $\delta \rightarrow 0$ in probability, both bounds converge to 1, giving Eq. (32). The continuous-oracle claim follows by combining $R_n(x_n^*) = \inf_{\mathcal{G}_n^+} R_n = (1 + o(1)) \inf_{[0, n]} R_n$ with Eq. (32). $\square$

Remark 2.14 (Verification of Eq. (31)). The conditions for Eq. (31) reduce to: (i) $\hat{a}/a \xrightarrow{P} 1$, by sample variance of plug-in influence values; (ii) $\hat{v}_n/v_n \xrightarrow{P} 1$, by sample variance of the synthetic estimator across multiple synthetic batches; and (iii) $\hat{b}^2(x)/b^2(x) \xrightarrow{P} 1$ uniformly over $\mathcal{G}_n^+$. Item (iii) is the binding requirement; for finite grids whose size grows polylogarithmically with n , the sample-split moment estimator achieves uniform relative consistency provided $b^2(x) \gg V_R(x) \vee v_n$ on the grid (i.e. the bias is detectable above the noise floor). On the bias-dominated portion of the grid this holds; on the variance-dominated portion, $\widehat{R}_n(x)$ converges to $V_R(x)$ and the harmonic-mean structure absorbs the noise in $\hat{b}^2$. The truncation $[\cdot]_+$ in step 3(c) introduces a positive bias that is also $o(V_R(x))$ in this regime. A formal proof uses Bernstein's inequality and a union bound over $\mathcal{G}_n^+$; details are deferred to an appendix.

2.7 Asymptotic distribution and inference

Point estimation and inference are different problems: the MSE-optimal estimator can have residual bias of the same order as its standard error, in which case Wald intervals undercover. The next result characterizes the asymptotic distribution of the linear estimator at the oracle allocation.

Proposition 2.15 (Centered asymptotic normality). *Suppose Assumptions 1.1 to 1.4 hold with $v_n > 0$, and that the conditional Lindeberg condition holds for ψ and ϕ_x . Define $\Omega_n(x) := \alpha^*(x)^2 V_R(x) + (1 - \alpha^*(x))^2 v_n$ as the stochastic-variance contribution to $\hat{\theta}^L(\alpha^*(x_n^*), x_n^*)$. Then*

$$\frac{\hat{\theta}^L(\alpha^*(x_n^*), x_n^*) - \theta^* - (1 - \alpha^*(x_n^*))\Delta(x_n^*)}{\sqrt{\Omega_n(x_n^*)}} \Rightarrow N(0, 1), \quad (33)$$

conditionally on $\mathcal{F}_C$ and unconditionally. Replacing oracle quantities by their estimates from Theorem 2.13 preserves the CLT under the same regularity.

Proof. By Lemma 1.4(i) and Assumptions 1.1 and 1.2, conditionally on $\mathcal{F}_C$,

$$\hat{\theta}^L - \theta^* = \alpha^*(\hat{\theta}_R - \theta^*) + (1 - \alpha^*)(\hat{\theta}_S - \theta_x) + (1 - \alpha^*)\Delta(x).$$

The first two summands are independent zero-mean martingale-difference averages with variances $\alpha^{*2}V_R$ and $(1 - \alpha^*)^2 v_n + o(v_n)$ respectively; the third is $\mathcal{F}_C$ -measurable. The Lindeberg-Feller CLT applied to the sum of two independent triangular arrays gives conditional asymptotic normality of the centered numerator with limiting variance $\Omega_n(x_n^*)$. Adaptive oracle equivalence (Theorem 2.13) and Slutsky's theorem give the result with estimated weights. $\square$

Corollary 2.16 (Wald inference around θ^*). *A two-sided Wald interval centered at $\hat{\theta}^L$ with width $z_{1-\alpha/2}\sqrt{\Omega_n(x_n^*)}$ has asymptotic coverage $1 - \alpha$ for θ^* if and only if*

$$(1 - \alpha^*(x_n^*))\Delta(x_n^*) = o_p(\sqrt{\Omega_n(x_n^*)}). \quad (34)$$

Under Theorem 2.11, in Regime III with $\rho = 1$ and $\beta > 1/2$, Eq. (34) holds; in Regime V with $\beta = 1/2$ and $\rho = 1$, it generically fails.

Proof. The first claim is immediate from Proposition 2.15. For Regime III with $\rho = 1$, $\beta > 1/2$: $\Omega_n \asymp n^{-1}$, $\Delta(x_n^*)^2$ has expectation $b^2(x_n^*) \asymp n^{-4\beta/(2\beta+1)}$, and $4\beta/(2\beta+1) > 1$ iff $\beta > 1/2$. So $\Delta(x_n^*) = O_p(n^{-2\beta/(2\beta+1)}) = o(n^{-1/2})$. Multiplying by $1 - \alpha^* = O(1)$ preserves the order, giving Eq. (34).

For Regime V with $\rho = 1$ and $a > c$: $\Omega_n \asymp n^{-1}$, $b^2(x_n^*) \asymp n^{-2\beta} = n^{-1}$, so $\Delta(x_n^*)$ is the same order as $\sqrt{\Omega_n}$, and Eq. (34) fails generically. $\square$

Remark 2.17 (Bias-aware inference). When Eq. (34) fails, valid intervals can be obtained by (i) undersmoothing (allocate $x > x_n^*$ so that $b^2(x) = o(V_R(x) + v_n)$, paying a constant-factor MSE penalty), (ii) explicit bias subtraction using validation data (the PPI construction of Section 3), or (iii) honest bias-aware intervals that widen by a bound on $|\Delta(x_n^*)|$. We adopt (ii) in the next section.

3 Theory II: Validation-Debiased (PPI) Estimator

Throughout this section we work with the PPI configuration: $V_x = [n] \setminus C_x$, $E_x = \emptyset$, $n_v := |V_x| = n - x$, and $\hat{\theta}^{\text{PP}}(x)$ defined by Eq. (15). We restrict to mean-functional estimands $\theta^* = \mathbb{E}_{P^*}[g(W)]$ and assume the synthetic generator preserves the marginal distribution of X in the sense of Eq. (14).

3.1 Decomposition of the PPI error

Lemma 3.1 (PPI error decomposition). *Under Assumption 1.2 and the conditional independence of Lemma 1.4(ii),*

$$\hat{\theta}^{\text{PP}}(x) - \theta^* = [\hat{\theta}_S(x) - \theta_x] - [\hat{\theta}_S^V(x) - \theta_x] + [\hat{\theta}_R^V - \theta^*], \quad (35)$$

where the three summands have zero conditional mean given $\mathcal{F}_C$ and are conditionally pairwise uncorrelated, except that the last two are negatively correlated through shared validation features.

Proof. Using Eq. (15), $\hat{\theta}^{\text{PP}}(x) - \theta^* = \hat{\theta}_S(x) - \hat{\theta}_S^V(x) + \hat{\theta}_R^V - \theta^*$. Adding and subtracting θ_x to the synthetic terms gives Eq. (35). The conditional means: $\mathbb{E}[\hat{\theta}_S(x) - \theta_x \mid \mathcal{F}_C] = 0$ by Assumption 1.2; $\mathbb{E}[\hat{\theta}_S^V(x) - \theta_x \mid \mathcal{F}_C] = \mathbb{E}[n_v^{-1} \sum_{V_x} (\tilde{g}_x(X_i) - \theta_x) \mid \mathcal{F}_C] = 0$ by Eq. (14) and the disjointness $V_x \cap C_x = \emptyset$; $\mathbb{E}[\hat{\theta}_R^V - \theta^*] = 0$. Conditional uncorrelatedness of the first summand with the others follows from Lemma 1.4(ii). $\square$

Lemma 3.2 (Working MSE for PPI). *Define the residual function $\rho_x : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ by $\rho_x(W) := g(W) - \tilde{g}_x(X)$ and the conditional residual variance*

$$\sigma_{\text{ppi}}^2(x) := \text{Var}_{P^*}(\rho_x(W) \mid \mathcal{F}_C). \quad (36)$$

Then

$$\mathbb{E}[(\hat{\theta}^{\text{PP}}(x) - \theta^*)^2] = v_n + \frac{\mathbb{E}[\sigma_{\text{ppi}}^2(x)]}{n_v} + o(v_n + \mathbb{E}[\sigma_{\text{ppi}}^2(x)]/n_v). \quad (37)$$

Proof. By Lemma 3.1, $\hat{\theta}^{\text{PP}}(x) - \theta^* = A + B + C$ with $A = \hat{\theta}_S(x) - \theta_x$, $B = -(\hat{\theta}_S^V(x) - \theta_x)$, and $C = \hat{\theta}_R^V - \theta^*$. We have $\mathbb{E}[A \mid \mathcal{F}_C] = 0$ by Assumption 1.2, $\mathbb{E}[B \mid \mathcal{F}_C] = -\mathbb{E}_{X \sim P^*}[\tilde{g}_x(X) - \theta_x \mid \mathcal{F}_C] = 0$ by Eq. (14), and $\mathbb{E}[C \mid \mathcal{F}_C] = \mathbb{E}_{P^*}[g(W)] - \theta^* = 0$ since $\{W_i\}_{i \in V_x} \perp\!\!\!\perp \mathcal{F}_C$. Hence the conditional mean is zero and the conditional MSE equals the conditional variance.

By Lemma 1.4(ii), $A \perp\!\!\!\perp (B, C) \mid \mathcal{F}_C$, so $\text{Cov}(A, B + C \mid \mathcal{F}_C) = 0$ and therefore

$$\mathbb{E}[(\hat{\theta}^{\text{PP}}(x) - \theta^*)^2 \mid \mathcal{F}_C] = \text{Var}(A \mid \mathcal{F}_C) + \text{Var}(B + C \mid \mathcal{F}_C).$$

For the first term, $\text{Var}(A \mid \mathcal{F}_C) = \sigma_S^2(x)/m_n + o(m_n^{-1})$ by Assumption 1.2. For the second, write

$$B + C = \hat{\theta}_R^V - \hat{\theta}_S^V(x) = \frac{1}{n_v} \sum_{i \in V_x} [g(W_i) - \tilde{g}_x(X_i)] = \frac{1}{n_v} \sum_{i \in V_x} \rho_x(W_i).$$

Conditionally on $\mathcal{F}_C$, the $\rho_x(W_i)$ are i.i.d. across $i \in V_x$ with variance $\sigma_{\text{ppi}}^2(x)$ (the conditional mean $\mathbb{E}[\rho_x(W) \mid \mathcal{F}_C] = -\Delta(x)$ enters $B + C$ only through the empirical-mean recentering, which leaves the variance unchanged). Hence $\text{Var}(B + C \mid \mathcal{F}_C) = \sigma_{\text{ppi}}^2(x)/n_v$.

Combining,

$$\mathbb{E}[(\hat{\theta}^{\text{PP}}(x) - \theta^*)^2 \mid \mathcal{F}_C] = \frac{\sigma_S^2(x)}{m_n} + \frac{\sigma_{\text{ppi}}^2(x)}{n_v} + o(m_n^{-1}).$$

Taking unconditional expectations and applying Assumption 1.2 (so that $\mathbb{E}[\sigma_S^2(x)/m_n] = v_n\{1 + o(1)\}$) yields Eq. (37). $\square$

3.2 Working model for PPI: residual variance decay

The residual variance $\sigma_{\text{ppi}}^2(x)$ has a useful decomposition. Define $\tilde{g}^*(X) := \mathbb{E}_{P^*}[g(W) \mid X]$ (the true conditional mean function) and $\sigma_{\text{irr}}^2 := \mathbb{E}_{P^*}[\text{Var}_{P^*}(g(W) \mid X)]$ (the irreducible noise level).

Lemma 3.3 (Residual variance decomposition). *For every x and almost every realization of $\mathcal{F}_C$,*

$$\sigma_{\text{ppi}}^2(x) = \sigma_{\text{irr}}^2 + \text{Var}_{P^*}(\tilde{g}^*(X) - \tilde{g}_x(X) \mid \mathcal{F}_C). \quad (38)$$

Proof. Decompose $\rho_x(W) = g(W) - \tilde{g}_x(X) = [g(W) - \tilde{g}^*(X)] + [\tilde{g}^*(X) - \tilde{g}_x(X)]$. Conditional on $\mathcal{F}_C$, $\tilde{g}_x$ is fixed, so the second summand is a function of X alone. Conditioning further on X , $\mathbb{E}_{P^*}[g(W) - \tilde{g}^*(X) \mid X, \mathcal{F}_C] = \mathbb{E}_{P^*}[g(W) \mid X] - \tilde{g}^*(X) = 0$ by definition of $\tilde{g}^*$. Hence

$$\mathbb{E}_{P^*}[(g(W) - \tilde{g}^*(X))(\tilde{g}^*(X) - \tilde{g}_x(X)) \mid \mathcal{F}_C] = \mathbb{E}_{P^*}[(\tilde{g}^*(X) - \tilde{g}_x(X)) \mathbb{E}_{P^*}[g(W) - \tilde{g}^*(X) \mid X, \mathcal{F}_C] \mid \mathcal{F}_C] = 0,$$

i.e., the two summands of ρ_x are conditionally uncorrelated. Therefore

$$\text{Var}_{P^*}(\rho_x(W) \mid \mathcal{F}_C) = \text{Var}_{P^*}(g(W) - \tilde{g}^*(X) \mid \mathcal{F}_C) + \text{Var}_{P^*}(\tilde{g}^*(X) - \tilde{g}_x(X) \mid \mathcal{F}_C).$$

The first term equals $\mathbb{E}_{P^*}[\text{Var}_{P^*}(g(W) \mid X)] = \sigma_{\text{irr}}^2$ (independent of $\mathcal{F}_C$ since $g(W) - \tilde{g}^*(X)$ has X -conditional mean 0 and the X -marginal is P^*). $\square$

Assumption 3.1 (Power-law residual variance). There exist $d > 0$, $\gamma \in (0, \beta]$, and a deterministic sequence $\mu_x \rightarrow 0$ such that

$$\mathbb{E}[\text{Var}_{P^*}(\tilde{g}^*(X) - \tilde{g}_x(X) \mid \mathcal{F}_C)] = d x^{-2\gamma} (1 + \mu_x). \quad (39)$$

The constraint $\gamma \leq \beta$ is automatic: since $\Delta(x) = \mathbb{E}_{P^*}[\tilde{g}_x(X) \mid \mathcal{F}_C] - \theta^*$, by Jensen's inequality $\mathbb{E}[\Delta(x)^2] \leq \mathbb{E}[\mathbb{E}_{P^*}[(\tilde{g}_x(X) - \tilde{g}^*(X))^2 \mid \mathcal{F}_C]] + (\mathbb{E}[\Delta(x)])^2$, and absorbing constants gives $b^2(x) \lesssim \mathbb{E} \text{Var}_{P^*}(\tilde{g}_x(X) - \tilde{g}^*(X)) + \text{const}$, hence $\beta \geq \gamma$ asymptotically.

Under Assumption 3.1,

$$\mathbb{E}[\sigma_{\text{ppi}}^2(x)] = \sigma_{\text{irr}}^2 + d x^{-2\gamma} (1 + \mu_x). \quad (40)$$

Substituting in Eq. (37) gives the working PPI MSE

$$\Psi_n(x) := v_n + \frac{\sigma_{\text{irr}}^2 + d x^{-2\gamma}}{n - x}, \quad x \in (0, n). \quad (41)$$

3.3 First-order condition and phase diagram for PPI

Proposition 3.4 (PPI first-order condition). *Under Assumptions 1.2, 1.4 and 3.1, every interior critical point $x \in (0, n)$ of Ψ_n satisfies*

$$\boxed{\sigma_{\text{irr}}^2 x^{2\gamma+1} + (2\gamma + 1) dx = 2\gamma dn.} \quad (42)$$

For every $\gamma > 0$, $\sigma_{\text{irr}}^2, d > 0$, the LHS is strictly increasing on $(0, \infty)$, so Eq. (42) has a unique solution $x_n^{\text{PP}} > 0$.

Proof. Differentiate Ψ_n :

$$\Psi'_n(x) = \frac{-2\gamma dx^{-2\gamma-1}(n-x) + (\sigma_{\text{irr}}^2 + dx^{-2\gamma})}{(n-x)^2}.$$

Setting the numerator to zero: $2\gamma dx^{-2\gamma-1}(n-x) = \sigma_{\text{irr}}^2 + dx^{-2\gamma}$. Multiplying by $x^{2\gamma+1}$ and rearranging gives Eq. (42). The LHS $f(x) := \sigma_{\text{irr}}^2 x^{2\gamma+1} + (2\gamma + 1)dx$ is strictly increasing with $f(0) = 0$ and $f(\infty) = \infty$, so the equation has a unique positive solution. $\square$

Theorem 3.5 (PPI phase diagram). *Under Assumptions 1.2, 1.4 and 3.1 with $\sigma_{\text{irr}}^2 > 0$, $d > 0$, $\gamma > 0$:*

(i) *The unique solution of Eq. (42) is*

$$x_n^{\text{PP}} = \left(\frac{2\gamma d}{\sigma_{\text{irr}}^2} \right)^{1/(2\gamma+1)} n^{1/(2\gamma+1)} \{1 + o(1)\}. \quad (43)$$

In particular $x_n^{\text{PP}}/n \rightarrow 0$ for every $\gamma > 0$.

(ii) *The optimal PPI risk satisfies*

$$\Psi_n(x_n^{\text{PP}}) = v_n + \frac{\sigma_{\text{irr}}^2}{n} + O(n^{-(4\gamma+1)/(2\gamma+1)}). \quad (44)$$

(iii) *In the headline regime $\rho = 1$:*

$$\Psi_n(x_n^{\text{PP}}) = \frac{\sigma_{\text{irr}}^2 + v_0}{n} \{1 + o(1)\}.$$

For $\rho > 1$, $v_n = o(1/n)$ so $\Psi_n(x_n^{\text{PP}}) \sim \sigma_{\text{irr}}^2/n$. For $\rho < 1$, v_n dominates: $\Psi_n(x_n^{\text{PP}}) \sim v_0 n^{-\rho}$.

(iv) *PPI dominates the all-real estimator (rate a/n where $a = \sigma_{\text{irr}}^2 + \text{Var}_{P^*}(\tilde{g}^*(X))$) asymptotically iff $\sigma_{\text{irr}}^2 + v_n \cdot n < a$ in the limit. With $\rho \geq 1$ this reduces to $\sigma_{\text{irr}}^2 < a$, equivalently $\text{Var}_{P^*}(\tilde{g}^*(X)) > 0$ (the regression has nonzero predictive power).*

Proof. (i) Conjecture $x_n^{\text{PP}} \rightarrow \infty$ with $x_n^{\text{PP}}/n \rightarrow 0$. Then in Eq. (42), the LHS term $(2\gamma + 1)dx_n^{\text{PP}}$ is $o(n)$ if $x_n^{\text{PP}} = o(n)$, while $\sigma_{\text{irr}}^2 x_n^{\text{PP} 2\gamma+1} = 2\gamma dn\{1 + o(1)\}$. Solving: $x_n^{\text{PP}} = (2\gamma d/\sigma_{\text{irr}}^2)^{1/(2\gamma+1)} n^{1/(2\gamma+1)} \{1 + o(1)\}$. Verify the conjecture: $1/(2\gamma + 1) < 1$ for $\gamma > 0$, so $x_n^{\text{PP}} = o(n)$, and the dropped term $(2\gamma + 1)dx_n^{\text{PP}} = O(n^{1/(2\gamma+1)}) = o(n)$. Hence the asymptotic equation is consistent.

(ii) At x_n^{PP} , $dx_n^{\text{PP} - 2\gamma} = d(2\gamma d/\sigma_{\text{irr}}^2)^{-2\gamma/(2\gamma+1)} n^{-2\gamma/(2\gamma+1)} = O(n^{-2\gamma/(2\gamma+1)})$, while σ_{irr}^2 is constant. So $\sigma_{\text{irr}}^2 + dx_n^{\text{PP} - 2\gamma} = \sigma_{\text{irr}}^2 \{1 + o(1)\}$. And $n - x_n^{\text{PP}} = n\{1 + o(1)\}$. Substituting into Eq. (41):

$$\Psi_n(x_n^{\text{PP}}) = v_n + \frac{\sigma_{\text{irr}}^2 \{1 + o(1)\}}{n\{1 + o(1)\}} = v_n + \sigma_{\text{irr}}^2/n + O(n^{-1} \cdot n^{-2\gamma/(2\gamma+1)}) = v_n + \sigma_{\text{irr}}^2/n + O(n^{-(4\gamma+1)/(2\gamma+1)}).$$

The error term is $o(1/n)$ since $(4\gamma + 1)/(2\gamma + 1) > 1$ iff $\gamma > 0$.

(iii) Direct from (ii) and Assumption 1.4.

(iv) The all-real estimator has variance a/n with $a = \text{Var}_{P^*}(g(W)) = \sigma_{\text{irr}}^2 + \text{Var}_{P^*}(\tilde{g}^*(X))$ by the law of total variance. Comparing to Eq. (44) establishes the claim. $\square$

Remark 3.6 (PPI is dimension-stable). The single-FOC, single-root structure of Proposition 3.4 contrasts sharply with the linear-estimator critical-point analysis (Lemma 2.6), which required separate treatment of $\beta \leq 1/2$ and $\beta > 1/2$. The reason is that the PPI variance $\Psi_n(x)$ is convex in x on $(0, n)$, while the linear profiled risk $R_n(x)$ is not generically convex. Convexity of Ψ_n comes from the bias-correction structure: the residual $\sigma_{\text{irr}}^2 + dx^{-2\gamma}$ is a positive convex function divided by $n - x$, which preserves quasi-convexity by elementary calculation.

Remark 3.7 (Plateau bias for PPI). Replacing Assumption 3.1 by $\mathbb{E}[\text{Var}_{P^*}(\tilde{g}^*(X) - \tilde{g}_x(X) \mid \mathcal{F}_C)] = d_\infty^2 + dx^{-2\gamma}\{1 + o(1)\}$ with $d_\infty^2 > 0$ (irreducible regression error from generator misspecification) yields $\mathbb{E}[\sigma_{\text{ppi}}^2(x)] \geq \sigma_{\text{irr}}^2 + d_\infty^2$, and the FOC becomes $(\sigma_{\text{irr}}^2 + d_\infty^2)x^{2\gamma+1} + (2\gamma + 1)dx = 2\gamma dn$, with the same scaling $x_n^{\text{PP}} \asymp n^{1/(2\gamma+1)}$ but a worse constant. The leading PPI risk becomes $v_n + (\sigma_{\text{irr}}^2 + d_\infty^2)/n$, still $O(1/n)$. *PPI is robust to plateau-type misspecification: the rate is preserved, only the constant degrades.* This contrasts with the linear estimator, where plateau bias caps calibration at $O(1)$ regardless of β .

3.4 Asymptotic distribution and inference for PPI

Proposition 3.8 (PPI is asymptotically normal and unbiased). *Under Assumptions 1.2 and 3.1 and the conditional Lindeberg condition for ρ_x , with $x = x_n^{\text{PP}}$ from Eq. (43),*

$$\frac{\hat{\theta}^{\text{PP}}(x_n^{\text{PP}}) - \theta^*}{\sqrt{\Psi_n(x_n^{\text{PP}})}} \Rightarrow N(0, 1). \quad (45)$$

A two-sided Wald interval centered at $\hat{\theta}^{\text{PP}}$ with width $z_{1-\alpha/2}\sqrt{\hat{\Psi}_n}$, where $\hat{\Psi}_n = \hat{v}_n + n_v^{-2} \sum_{i \in V_x} (\rho_{\hat{g}_x}(W_i) - \bar{\rho})^2$, has asymptotic coverage $1 - \alpha$ for θ^* without any bias-rate conditions.

Proof. By Lemma 3.1, $\hat{\theta}^{\text{PP}} - \theta^* = A + (B + C)$ with three zero-conditional-mean summands; $A \perp\!\!\!\perp (B + C) \mid \mathcal{F}_C$. Each is an i.i.d. sum of mean-zero random variables (synthetic and real, respectively). Apply Lindeberg-Feller conditionally on $\mathcal{F}_C$ to obtain conditional asymptotic normality of $\sqrt{\Psi_n}^{-1}(A + B + C) \Rightarrow N(0, 1)$. Unconditional convergence follows by dominated convergence after verifying $\mathbb{E}[\Psi_n^{-1}(\hat{\theta}^{\text{PP}} - \theta^*)^2 \mid \mathcal{F}_C] \rightarrow 1$ in probability, which holds under Assumption 3.1. The plug-in variance estimator $\hat{\Psi}_n$ is consistent under standard conditions on the sample analogues. The absence of bias-rate conditions follows because $\mathbb{E}[\hat{\theta}^{\text{PP}} - \theta^*] = 0$ exactly (under Eq. (14)); see Lemma 3.1. $\square$

4 Comparison and Decision Rule

We collect the asymptotic risks of the three candidate estimators in the headline regime $\rho = 1$. Let $a = \sigma_{\text{irr}}^2 + \sigma_{\text{red}}^2$ where $\sigma_{\text{red}}^2 := \text{Var}_{P^*}(\tilde{g}^*(X))$ is the explainable variance.

Theorem 4.1 (Headline-regime comparison). *Under Assumptions 1.1 to 1.4 and 3.1 with $\rho = 1$:*

(i) *All-real estimator:* $\mathbb{E}[(\hat{\theta}_R - \theta^*)^2] = a/n + o(1/n)$.

(ii) *Linear-allocation estimator at oracle:*

$$R_n(x_n^*) = \begin{cases} \frac{a}{n} - \Theta(n^{-2}) & \beta < 1/2, \\ \frac{av_0}{(a + v_0)n} \{1 + o(1)\} & \beta > 1/2. \end{cases}$$

(iii) *PPI estimator at oracle:* $\Psi_n(x_n^{\text{PP}}) = (\sigma_{\text{irr}}^2 + v_0)/n \{1 + o(1)\}$ for every $\gamma > 0$.

In particular, when $\beta > 1/2$, the linear estimator strictly dominates PPI on first-order MSE:

$$\frac{R_n(x_n^*)}{\Psi_n(x_n^{\text{PP}})} \rightarrow \frac{av_0/(a+v_0)}{\sigma_{\text{irr}}^2 + v_0} < 1.$$

When $\beta < 1/2$, the linear estimator achieves at most a constant-factor improvement over a/n , while PPI achieves the strictly better rate $\min(\sigma_{\text{irr}}^2 + v_0, a)/n$, which is strictly smaller iff $\sigma_{\text{red}}^2 > v_0$.

Proof. (i)–(iii) restate Theorem 2.11 (Regimes II, III) and Theorem 3.5(iii). For the comparison: $av_0/(a+v_0) < \min(a, v_0)$, the harmonic mean is strictly smaller than the smaller of the two arguments, while $\sigma_{\text{irr}}^2 + v_0 \geq v_0 \geq \min(a, v_0)$, hence $\sigma_{\text{irr}}^2 + v_0 > av_0/(a+v_0)$ whenever $\sigma_{\text{irr}}^2, v_0 > 0$.

For Regime II: linear MSE is $a/n - \Theta(n^{-2})$, asymptotic constant a . PPI MSE constant is $\sigma_{\text{irr}}^2 + v_0$. PPI smaller iff $\sigma_{\text{irr}}^2 + v_0 < a = \sigma_{\text{irr}}^2 + \sigma_{\text{red}}^2$, i.e., $v_0 < \sigma_{\text{red}}^2$. $\square$

Theorem 4.1 translates into the following decision rule.

Corollary 4.2 (When to use which estimator). *Under the regularity of Theorem 4.1:*

- If $\beta > 1/2$ and inference is not required, use the linear estimator: it achieves a strictly better first-order MSE constant than PPI, and Corollary 2.16 shows that bias is negligible relative to standard error so Wald inference is valid anyway.
- If $\beta \leq 1/2$ and the regression has nonzero predictive power ($\sigma_{\text{red}}^2 > v_0$), use PPI: it achieves a faster first-order rate than linear (which is capped at a/n), and inference is automatically valid.
- If valid coverage is the primary goal regardless of regime, use PPI: it is always asymptotically unbiased.
- If plateau-type misspecification is suspected, use PPI: it is rate-robust (Remark 3.7), while linear is not (Proposition 2.12).

Remark 4.3 (Combined estimator). A natural extension is the linear combination of the linear and PPI estimators with optimally chosen mixing weight. The combined risk equals the harmonic mean of the two component risks (under independence enforced by additional sample splitting), inheriting the better of the two rates. In the headline regime with $\beta > 1/2$, the linear component dominates, so the combined estimator coincides with the linear estimator at first order. We omit details.